Question ID: c39dbbdf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

Line r is defined by the equation $4x - 9y = 3$. Line s is parallel to line r in the *xy*-plane. What is the slope of line s?

Answer

A. $\frac{9}{4}$

B. $\frac{4}{9}$

C. -4

D. -9

Correct Answer: B

Rationale

Choice B is correct. It's given that line s is parallel to line r in the *xy*-plane. This means that the slope of line r is equal to the slope of line s. Line r is defined by the equation $4x - 9y = 3$. This equation can be written in slope-intercept form $y = mx + b$, where m represents the slope of the line and b represents the *y*-coordinate of the *y*-intercept of the line. Subtracting $4x$ from both sides of the equation $4x - 9y = 3$ yields $-9y = -4x + 3$. Dividing both sides of this equation by -9 yields $y = \frac{4}{9}x - \frac{1}{3}$. Therefore, the slope of line r is $\frac{4}{9}$. Since parallel lines have equal slopes, the slope of line s is also $\frac{4}{9}$.

Choice A is incorrect. This is the reciprocal of the slope of line s, not the slope of line s.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.